

FULTON COUNTY AP, IN, USA

WMO: 720736

Lat: 41.066N

Lon: 86.182W

Elev: 791

StdP: 14.28

Time Zone: -5.00 (NAE)

Period: 10-19

WBAN: 00265

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.5	8.6	-6.5	4.0	4.2	-0.5	5.5	9.1	23.3	33.7	20.8	27.0	5.8	250	0.472

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.4	91.4	74.2	90.2	73.7	88.0	72.7	78.2	87.3	76.8	85.7	75.2	83.4	8.9	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.4	137.1	81.5	73.3	127.6	80.4	72.6	124.5	79.7	42.2	87.9	40.6	86.4	39.3	83.9	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.4	18.2	16.3	-7.6	97.0	5.6	3.4	-11.6	99.5	-14.9	101.5	-18.1	103.4	-22.1	105.8		
			-8.2	80.1	5.6	1.5	-12.2	81.2	-15.5	82.1	-18.6	83.0	-22.7	84.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.2	27.8	30.8	40.4	51.5	64.6	72.5	75.9	73.6	67.6	55.4	42.4	34.3
	DBStd	19.16	11.41	11.89	10.90	8.46	9.38	6.04	5.94	4.93	7.99	8.87	9.33	9.97
	HDD50	2462	690	546	339	80	5	0	0	0	0	42	265	495
	HDD65	5448	1154	958	765	410	127	12	2	3	62	323	680	952
	CDD50	3627	1	8	43	125	459	676	804	731	528	209	35	8
	CDD65	1137	0	0	3	5	116	238	341	269	140	24	1	0
	CDH74	10498	0	0	25	72	1094	2045	3457	2340	1249	213	3	0
	CDH80	3674	0	0	2	5	353	710	1388	745	426	45	0	0
Wind	WSAvg	7.0	8.5	8.3	8.3	8.8	7.1	6.0	4.8	4.4	5.3	6.9	7.8	7.8
Precipitation	PrecAvg	38.3	2.0	1.7	2.6	3.8	4.1	4.0	4.2	3.6	3.5	2.8	3.3	2.7
	PrecMax	62.1	6.1	6.0	5.5	8.2	7.3	9.6	9.6	12.4	9.9	9.0	9.1	8.0
	PrecMin	27.0	0.0	0.0	0.4	0.5	0.8	1.3	0.7	0.6	0.0	0.3	1.0	0.3
	PrecStd	6.8	1.4	1.2	1.1	1.7	1.7	1.8	2.4	2.3	2.2	1.8	1.7	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.4	64.2	79.4	80.9	92.7	94.8	98.7	91.4	95.0	85.9	72.5	63.3
		MCWB	55.2	55.2	64.1	61.5	68.5	73.7	76.5	74.7	73.7	69.5	59.9	58.2
	2%	DB	53.7	59.4	70.0	76.5	87.7	90.5	92.7	89.9	89.9	80.5	65.9	58.6
		MCWB	50.6	53.6	58.2	60.6	68.5	72.4	75.8	74.5	73.3	65.6	58.3	54.3
	5%	DB	47.8	53.5	63.3	71.9	83.8	88.0	90.2	87.6	85.6	75.0	61.4	52.0
		MCWB	44.9	48.8	55.3	58.8	66.8	70.9	74.9	73.6	71.0	63.2	54.0	48.9
	10%	DB	43.0	45.6	58.2	66.5	80.8	84.1	87.9	84.0	81.2	70.4	55.5	46.2
		MCWB	39.5	41.1	50.5	55.7	65.7	69.9	73.9	72.2	68.0	61.6	49.6	43.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.5	59.6	64.2	66.5	73.4	77.9	80.7	80.4	77.2	71.7	64.1	60.6
		MCDB	58.0	62.7	77.5	75.4	83.7	88.3	89.8	83.6	88.3	81.0	69.0	62.3
	2%	WB	50.6	53.5	60.6	63.3	71.4	76.2	78.8	77.8	74.9	69.0	60.3	54.3
		MCDB	52.3	58.0	68.6	70.0	80.8	86.0	88.8	84.7	84.1	75.7	64.1	57.0
	5%	WB	44.8	47.2	56.4	60.6	69.5	74.1	77.2	76.0	73.4	66.2	55.1	48.4
		MCDB	46.4	52.7	61.6	68.2	78.7	82.6	86.8	83.3	81.6	72.0	60.6	51.2
	10%	WB	39.7	41.5	51.6	58.1	68.0	72.4	75.7	74.4	71.7	63.0	50.7	44.7
		MCDB	43.2	45.5	57.4	66.7	77.3	80.5	84.2	81.1	79.1	68.9	56.6	46.5
Mean Daily Temperature Range	5% DB	MDBR	13.6	14.5	16.4	20.2	20.0	19.0	19.4	19.7	20.3	19.3	15.2	11.8
		MCDBR	18.3	21.1	22.8	26.1	23.8	22.5	20.9	22.0	22.4	24.0	21.8	17.3
		MCWBR	15.4	16.9	14.9	15.3	10.4	9.0	8.4	9.4	8.7	12.1	14.6	14.7
	5% WB	MCDBR	16.7	20.4	20.7	22.6	19.3	17.8	18.3	16.3	18.4	19.2	19.6	16.0
		MCWBR	15.7	17.1	15.1	16.5	9.8	8.8	8.9	8.2	8.7	12.7	15.2	14.8
Clear-Sky Solar Irradiance	taub	0.295	0.324	0.346	0.407	0.444	0.436	0.449	0.450	0.417	0.355	0.318	0.299	
	taud	2.417	2.327	2.362	2.202	2.148	2.216	2.181	2.167	2.239	2.416	2.464	2.441	
	Ebn at Noon	274	280	286	274	266	267	262	258	259	266	263	262	
	Edh at Noon	25	32	35	44	48	45	46	45	39	30	24	23	
All-Sky Solar Radiation	RadAvg	506	775	1134	1478	1691	1924	1932	1702	1401	934	596	423	
	RadStd	49	91	93	132	156	123	107	97	114	91	76	45	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page